# ITA0612 – MACHINE LEARNING

# CO2-AT1- APPLICATION BASED QUESTIONS

Submitted By : Shanuusree M (192412319)

Submitted To : Dr. Shri Vindhya A

1. A Genetic Algorithm is applied to maximize the function f(x) = $x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

**Given**

Objective function:

$$f(x) = x^2$$

Initial population:

$$P = \{2,4,6,8\}$$

**Step 1: Calculate Fitness Values**

The fitness of each individual is calculated using: $f(x) = x^2$

**For x = 2**

$$f(2) = 2^2 = 4$$

**For x = 4**

$$f(4) = 4^2 = 16$$

**For x = 6**

$$f(6) = 6^2 = 36$$

**For x = 8**

$$f(8) = 8^2 = 64$$

**Fitness Table**

| **Individual (x)** | **Fitness $f(x)$** |
|---|---|
| 2 | 4 |
| 4 | 16 |
| 6 | 36 |

**Step 2: Select Best Candidates**

The individuals with the highest fitness are selected.

| Rank | Individual | Fitness |
|---|---|---|
| 1 | 8 | 64 |
| 2 | 6 | 36 |
| 3 | 4 | 16 |
| 4 | 2 | 4 |

Selected parents:

$$P_1 = 8, P_2 = 6$$

**Step 3: Convert Selected Parents into Binary**

$$8 = (1000)_2$$
$$6 = (0110)_2$$

Therefore,

| Parent | Binary Form |
|---|---|
| 8 | 1000 |
| 6 | 0110 |

**Step 4: Perform Crossover**

Exchange the least significant bit (LSB).

**Before Crossover**

Parent 1: $100$

Parent 2: $0110$

LSB of Parent 1 = 0

LSB of Parent 2 = 0

**After Swapping LSBs**

Offspring 1: $1000$

Offspring 2: $0110$

Since both LSBs are the same, the offspring remain unchanged.

**Step 5: Convert Offspring to Decimal**

$$1000_2 = 8$$
$$0110_2 = 6$$

**Offspring Table**

| Offspring | Binary | Decimal |
|---|---|---|
| $O_1$ | 1000 | 8 |
| $O_2$ | 0110 | 6 |

**Step 6: Calculate Fitness of Offspring**

**For $O_1$ = 8**

$$f(8) = 8^2 = 64$$

**For $O_2$ = 6**

$$f(6) = 6^2 = 36$$

| Offspring | Fitness |
|---|---|
| 8 | 64 |
| 6 | 36 |

**Mutation in Genetic Algorithm**

Mutation randomly changes one or more bits in a chromosome.

**Example**

$$1000 \rightarrow 1001$$
$$8 \rightarrow 9$$

**Importance of Mutation**

1. Increases population diversity.
2. Prevents premature convergence.
3. Helps escape local optima.
4. Improves the chance of finding the global optimum solution.

2. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

**Given**

Advertisement exposure: $x_1 = 2$

Income level: $x_2 = 1$

Weights $w_1 = 0.5, w_2 = 0.3$

Bias: $b = -0.4$

**Step 1: Write the Perceptron Equation**

The net input is:

$$Net = w_1x_1 + w_2x_2 + b$$

**Step 2: Substitute Values**

$$Net = (0.5)(2) + (0.3)(1) + (-0.4)$$

**Step 3: Perform Multiplication**

$$Net = 1.0 + 0.3 - 0.4$$

**Step 4: Calculate Net Input**

$$Net = 0.9$$

Therefore,

$$\boxed{Net = 0.9}$$

**Step 5: Apply Step Activation Function**

$$y = \begin{cases} 1, & Net \geq 0 \\ 0, & Net < 0 \end{cases}$$

Since $0.9 \geq 0$ & $y = 1$

**Step 6: Interpret the Result**

Output: $\boxed{y = 1}$

where,

- 1= Purchase
- 0= No Purchase
- **Final Decision** $\boxed{\text{Customer will purchase the product}}$